

## SUMMARY TABLE: CHEMISTRY GRADE 10

### Sub-Strand 1.2: The Atom — Teacher Reference (Pre-filled)

| SUMMARY TABLE: CHEMISTRY GRADE 10 |                                                                                                                                                                            |
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| Sub-Strand                        | Sub-Strand 1.2: The Atom                                                                                                                                                   |
| Driving Question                  | DRIVING QUESTION: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                     | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                   | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>Lesson 1</b><br><b>The Mystery of Coloured Flames: Noticing and Wondering</b>                       | Students observed that sodium chloride produces a vivid yellow-orange flame, potassium permanganate produces a soft violet flame, and copper sulphate produces a blue-green flame, and that these colours are repeatable and specific to each substance.                                                                                                                                                                                                                                          | Students learned that different substances produce characteristic, repeatable colours in a flame, that this phenomenon is observable both in a school laboratory and in everyday Kenyan life such as at a cooking hearth, and that the colour is linked to the identity of the substance rather than to the colour of the solution or the temperature of the flame.                                                                 | Students cannot yet explain why different substances produce different colours — this productive gap is the engine of the entire unit. Initial ideas on the DQB reflect intuitive thinking rather than scientific models, and the initial atom drawings reveal baseline mental models that will be systematically developed across Lessons 2 through 9.                                                                                                                                                                                                                                               |
| <b>Lesson 2</b><br><b>What Is an Atom? Reviewing Atomic Structure — Dalton &amp; Rutherford Models</b> | Students observed, via the Rutherford Gold Foil simulation on the ARES platform, that most alpha particles passed straight through the gold foil with little or no deflection, a small number deflected at moderate angles, and a very small number bounced back at large angles. They also observed, through the Element Card Comparison activity, that an element's atomic number equals its proton count, and that neutron number is calculated by subtracting atomic number from mass number. | Students learned the definitions of atomic number, mass number, proton, neutron, and electron, and how to calculate sub-atomic particle counts. They learned the key features of the Rutherford nuclear model: an atom is mostly empty space, with a tiny, dense, positively charged nucleus at the centre and electrons somewhere in the surrounding space. They were introduced to the homestead-centre metaphor for the nucleus. | The Rutherford model explains why atoms of different elements have different proton counts — a fact that is the same as having different atomic numbers — but it cannot yet explain why different elements produce different flame colours. The model establishes that electrons exist outside the nucleus in mostly empty space, creating the central unresolved question for the unit: where exactly are those electrons, and how does their specific arrangement account for the precise colours seen at the Kenyan cooking hearth? This gap motivates Lesson 3 on Bohr's model and energy levels. |
| <b>Lesson 3</b>                                                                                        | Annotated Bohr model diagrams of sodium                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Bohr's planetary model proposes that                                                                                                                                                                                                                                                                                                                                                                                                | Bohr's model begins to explain the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| <b>Bohr's Model and Energy Levels</b>                                                             | <p>showing an electron in Shell 3 absorbing energy from a flame (Rift Valley cooking hearth), jumping to a higher energy level (excited state), then falling back to the ground state and releasing a yellow photon. Bohr model drawings for elements 1–10 (hydrogen to neon) showing electrons correctly distributed across shells (Shell 1: max 2e<sup>-</sup>; Shell 2: max 8e<sup>-</sup>). Electron arrangement notation for elements 11–15 written as comma-separated shell occupancies (e.g., Na = 2, 8, 1).</p>                                              | <p>electrons occupy fixed, discrete energy levels (shells) around the nucleus — they cannot exist between shells. Shell 1 holds a maximum of 2 electrons; Shell 2 holds a maximum of 8 electrons. When a flame supplies energy to an atom (e.g., sodium in cooking salt), the outermost electron absorbs that energy and jumps to a higher energy level — this is the excited state. The excited state is unstable; the electron immediately falls back to its original (ground state) energy level, releasing the absorbed energy as a photon of light. The colour of that light is determined by the exact amount of energy released, which depends on the specific spacing of the atom's energy levels. Because each element has uniquely spaced energy levels, each element produces a unique, characteristic colour — sodium always yellow, potassium always violet. This is why the cooking fire at the Rift Valley homestead changes colour depending on WHICH substance is added to the flame.</p> | <p>anchoring phenomenon: the reason different substances produce different flame colours is that different atoms have different electron energy level spacings. When thermal energy from the flame excites an electron to a higher level, the photon of light released as it falls back carries an amount of energy that is unique to that atom's energy level gap — producing a characteristic colour. Sodium's energy level spacing corresponds to yellow light (589 nm); potassium's spacing corresponds to violet light. This means flame colour is a direct observable fingerprint of atomic electron structure — connecting the culturally familiar cooking hearth observation to the internal quantum structure of atoms described by Bohr's model.</p> |
| <b>Lesson 4<br/>Isotopes and Relative Atomic Mass — Part 1: Meaning and Concept</b>               | <p>Groups completed Nuclear Composition Tables for C-12, C-14, <sup>1</sup>H, <sup>2</sup>H, <sup>3</sup>H, <sup>35</sup>Cl, and <sup>37</sup>Cl using AZX notation. During the Gallery Walk, learners used green and red sticky dots to identify what stays the same (proton number / atomic number) and what changes (neutron count / mass number) across isotopes of the same element. Learners also sorted 12 atom cards into isotope families, identifying sodium, potassium, and chlorine as elements from the anchoring Kenyan cooking hearth phenomenon.</p> | <p>Isotopes are atoms of the same element that have the same atomic number (same number of protons and electrons) but different mass numbers (different numbers of neutrons). Because atomic number is unchanged, isotopes share identical electron arrangements. Since flame colour is produced by electron energy transitions — not by nuclear mass — isotopes of the same element always produce the same flame colour. Carbon-14, used in dating Kenyan archaeological sites like Olorgesailie, is an isotope of carbon-12: same 6 protons, but 2 extra neutrons.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Isotopes of the same element produce the same colour in a flame test because flame colour is determined by the electron arrangement of an atom, which depends on atomic number (number of protons). Since isotopes share the same atomic number, their electrons occupy the same energy levels. When heated, those electrons jump to the same higher energy levels and release the same wavelengths of light when falling back — producing an identical colour. Mass number (and therefore neutron count) does not affect electron arrangement, so it does not affect flame colour.</p>                                                                                                                                                                     |
| <b>Lesson 5<br/>Isotopes and Relative Atomic Mass — Part 2: Calculation and Weighted Averages</b> | <p>Learners received isotopic abundance data cards for chlorine (Cl-35: 75%, Cl-37: 25%) and carbon (C-12: 98.9%, C-13: 1.1%). They observed that the periodic table lists chlorine's atomic mass as 35.5 — a value between the two whole-number isotope masses — and investigated why this non-</p>                                                                                                                                                                                                                                                                 | <p>Learners learned that relative atomic mass (RAM) is calculated as a weighted average of isotope masses, using the formula <math>RAM = \sum(\text{isotope mass} \times \text{fractional abundance})</math>. They calculated RAM for chlorine (<math>\approx 35.5</math>) and carbon (<math>\approx 12.01</math>), and verified their answers using a reasonableness check</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Learners explained that because all isotopes of an element share the same atomic number and therefore the same electron arrangement, they produce identical colours in a flame test. Cl-35 and Cl-37 would both produce the same flame colour because they both have 17 protons</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|                                                                                        | whole-number value appears.                                                                                                                                                                                                                                                                                                                                                                                                                                             | (RAM must fall between the lightest and heaviest isotope masses).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | and 17 electrons. Flame colour is a fingerprint of atomic number — not atomic mass. The Driving Question Board was updated to record: 'FLAME COLOUR = ATOMIC NUMBER fingerprint, NOT atomic mass.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Lesson 6</b><br><b>From Energy Levels to Orbitals: Introducing s and p Orbitals</b> | Students observed two ARES animations: (1) the limitation of Bohr's electron shells in explaining sub-level energy differences, and (2) the three-dimensional shapes of s orbitals (spherical) and p orbitals (dumbbell/figure-8 along x, y, z axes). Students also observed that sodium (Z=11) has its outermost electron in the 3s orbital and potassium (Z=19) has its outermost electron in the 4s orbital — two different orbitals at two different energy levels. | Students learned that within each principal energy level (shell), electrons occupy sub-regions called orbitals; that s orbitals are spherical and hold a maximum of 2 electrons; that there are three p orbitals (p <sub>x</sub> , p <sub>y</sub> , p <sub>z</sub> ) per principal level, each holding 2 electrons (6 electrons total in p sub-level); that orbitals fill in order 1s → 2s → 2p → 3s → 3p; and that the specific orbital holding an atom's outermost electron determines the precise energy gap — and therefore the precise colour of light — released when that electron returns from an excited state.                                                                   | Students explained the anchoring phenomenon more precisely than before: the sodium atom's lone outermost electron occupies the 3s orbital; when heated in a flame, it absorbs energy and jumps to a higher orbital, then releases exactly $3.37 \times 10^{-19}$ J as yellow-orange light (589 nm) when it falls back. Potassium's outermost electron in the 4s orbital has a different energy gap, so it releases a different quantum of energy, producing violet light. This is why different substances — salt at a Kenyan cooking hearth, or copper wire in the Kisumu school lab — always produce their own unique, repeatable flame colour: it is a direct fingerprint of orbital structure.                                                                   |
| <b>Lesson 7</b><br><b>Filling the Orbitals: Aufbau, Pauli, and Hund's Rules</b>        | Learners observed the 'Electron Seating Chart' manipulative activity in which arrow-cards (representing electrons with defined spins) were placed into orbital boxes for elements hydrogen through neon, revealing patterns in how electrons fill orbitals. Learners also observed a PhET simulation (or teacher-projected walkthrough) showing electrons being added to an atom one at a time, visually confirming the filling order.                                  | Learners learned the three rules that govern electron filling: (1) the Aufbau principle — electrons fill the lowest available energy orbital first (1s before 2s before 2p before 3s); (2) the Pauli exclusion principle — each orbital holds at most 2 electrons, and they must have opposite spins (one arrow up, one arrow down); (3) Hund's rule — when orbitals of equal energy are available (e.g., the three 2p orbitals), one electron enters each before any orbital is doubly occupied. Learners also learned to write electron configurations in s/p notation for elements 1–10 and extended this to sodium (1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup> 3s <sup>1</sup> ). | These three rules explain why sodium always produces the same yellow-orange colour in a flame: sodium's electron configuration always places its outermost electron in the 3s orbital (as determined by the Aufbau principle). When the flame provides energy, that specific 3s electron — and no other — absorbs it and jumps to a higher orbital. When it falls back to 3s, it releases exactly the energy corresponding to that orbital gap, which is the energy of yellow-orange light (589 nm). Because every sodium atom has this same orbital arrangement (enforced by the three filling rules), every sodium atom releases the same colour — explaining the consistent yellow flash seen when salt falls into the cooking fire at the Rift Valley homestead. |
| <b>Lesson 8</b><br><b>Electron Configuration for Elements 1–20 (s and p Notation)</b>  | Students observed the complete s and p electron configuration pattern for all 20 elements (H–Ca) on their colour-coded Configuration Table, noting that all Group 1 elements (H, Li, Na, K) share a lone                                                                                                                                                                                                                                                                | Students learned that the full s and p electron configuration for any element from H to Ca can be written systematically using the Aufbau order; that elements in the same group have identical outermost subshell                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Students explained that the yellow-orange flame of sodium and the violet flame of potassium at the Kenyan cooking hearth arise because both elements have one electron in an outermost s orbital, but                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|                                                                                                                            | outermost $ns^1$ electron, and that Group 2 elements (Be, Mg, Ca) share $ns^2$ , while halogens (F, Cl) share $ns^2np^5$ .                                                                                                                                                                                                               | types and electron counts; and that sodium ( $3s^1$ ) and potassium ( $4s^1$ ) differ only in the principal energy level of their lone outermost electron — which is why they produce similar but distinctly coloured flames.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | sodium's electron is in the third energy level and potassium's is in the fourth — meaning each releases a photon of a different energy (and therefore a different colour of visible light) when the excited electron falls back to its ground state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Lesson 9</b><br><b>Model Building, DQB Resolution, and Unit Consolidation:</b><br><b>Answering the Driving Question</b> | Students observed (via evidence station cards) flame test colour data for sodium, potassium, copper, lithium, and calcium; Rutherford gold-foil scattering diagrams; orbital filling diagrams for Na and K; and isotope abundance tables for chlorine — all gathered across Lessons 1–8 and reviewed in this lesson's evidence carousel. | Students learned that each atomic model (Dalton → Rutherford → Bohr → modern s/p orbital model) was driven to revision by specific experimental evidence; that sodium's lone $3s^1$ valence electron and potassium's lone $4s^1$ valence electron occupy orbitals at different energy levels, so when excited by flame heat they release photons of different energies corresponding to different colours (yellow at ~589 nm for Na; violet for K); that isotopes of the same element differ in neutron number and thus mass number, and that relative atomic mass is a weighted average of isotopic masses; and that the electron configurations of the first 20 elements can be written systematically using s and p notation. | Students explained the anchoring phenomenon — the yellow-orange glow when salt is spilled into the Rift Valley cooking fire and the violet flash from banana-peel ash — by constructing cumulative annotated models showing that sodium's outermost electron ( $3s^1$ ) and potassium's outermost electron ( $4s^1$ ) absorb thermal energy from the flame, jump to higher-energy excited states, and release that energy as photons of specific wavelengths when returning to ground state; because sodium's and potassium's valence electrons occupy orbitals at different energy levels, the energy — and therefore the colour — of the emitted photon is different for each element, making flame colour a direct fingerprint of atomic electron arrangement. |

## ATTRIBUTION AND LICENSING

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